

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

One protocol. Multiple CE paradigms. Measurable trade-offs.

Oleksii Furman · Łukasz Lenkiewicz · Marcel Musiałek · Maciej Zięba

Wrocław University of Science and Technology, Wrocław, Poland



genwro.AI



Code & project

Benchmark scope

18
datasets

14
methods

3
paradigms

2
backbones / task

5
folds

Source · Datasets · Methods · Full-Results

What is a counterfactual explanation?

A small input change that flips a model's prediction.

ILLUSTRATIVE LOAN MODEL

Original input

€3,000

Monthly income

Declined

↓ Change income only

Counterfactual input

€4,000

Monthly income

Approved

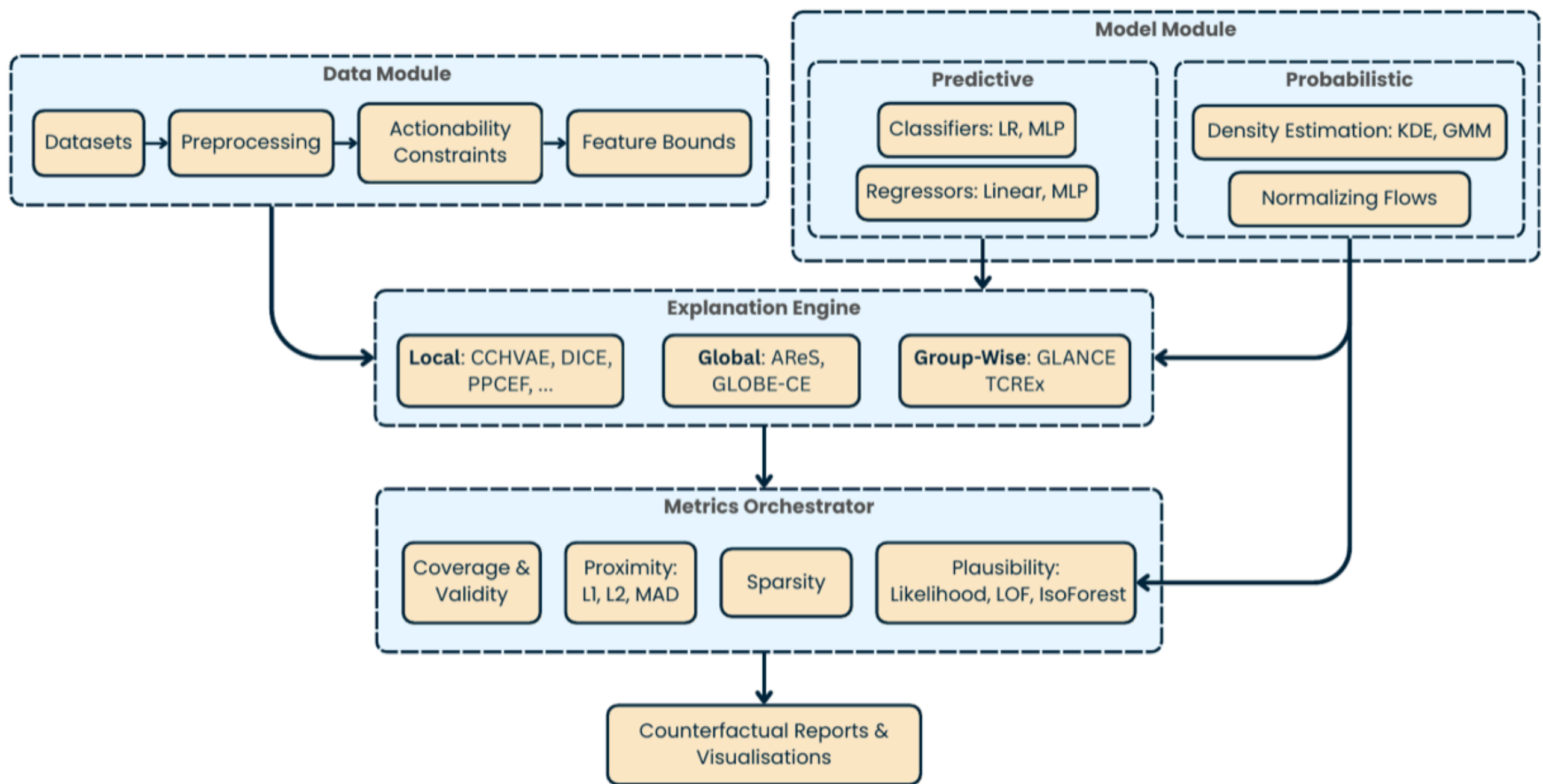
All other features unchanged.

Toy rule: approve at $\geq$ €4,000/month.

Toy model, not benchmark data.

Source · Introduction

One evaluation framework



SHARED EVALUATION PROTOCOL

Source · Introduction

Three contributions

CONTROL

Controlled evaluation protocol

COVER

Broad benchmark across CE paradigms

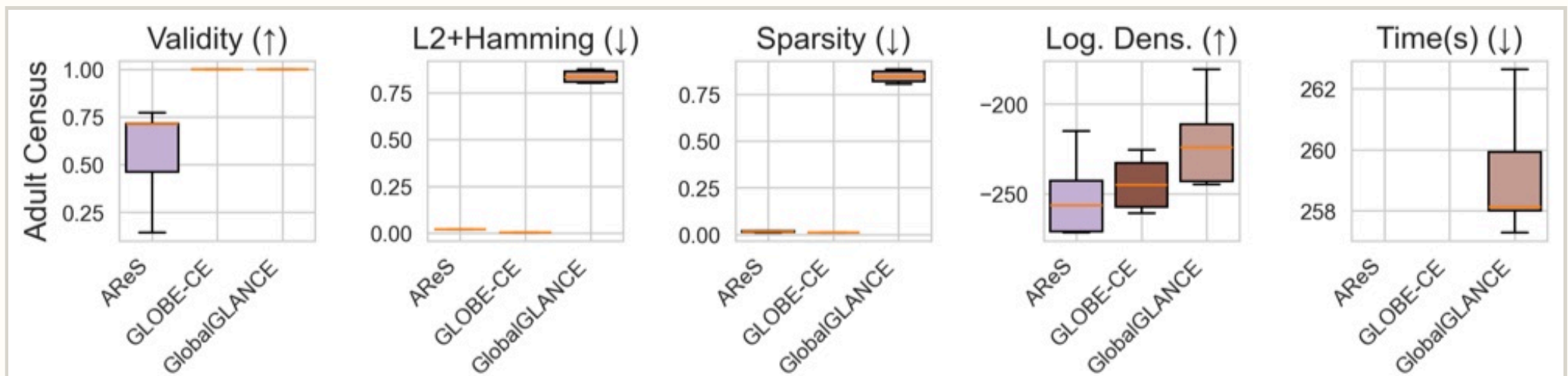
EXTEND

Extensible open-source benchmark workbench

Source · Introduction

Global: validity and failures

GLOBE-CE and GlobalGLANCE: stronger validity. AReS: frequent missing outputs.

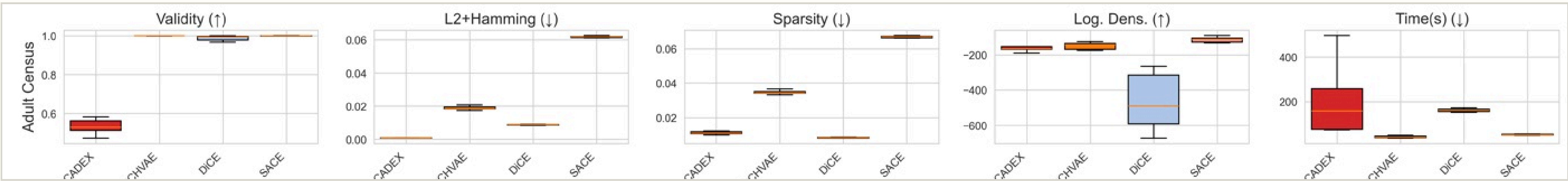


ADULT CENSUS · GLOBAL METHODS

Source · Results · tab:global_metrics_mlp

Local: no single winner

Validity, proximity, plausibility, and runtime trade off.



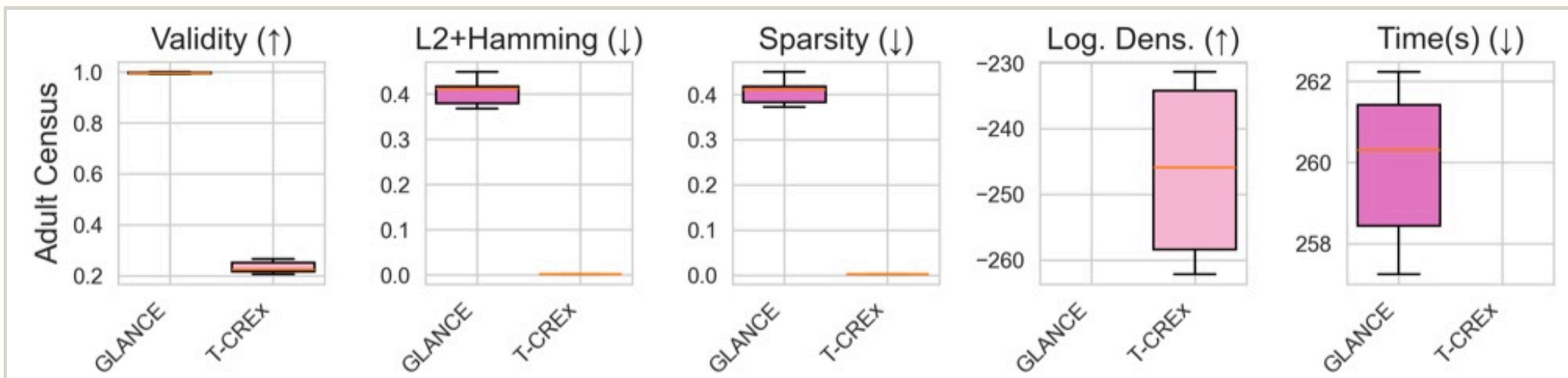
ADULT CENSUS · LOCAL METHODS

Sparsity direction unresolved; no ranking.

Source · Results · Conclusions · Metrics

Group-wise: reach versus change

GLANCE reaches more; T-CREx makes smaller, more plausible shifts when applicable.



ADULT CENSUS · GROUP-WISE METHODS

Source · Results · tab:group_metrics_mlp